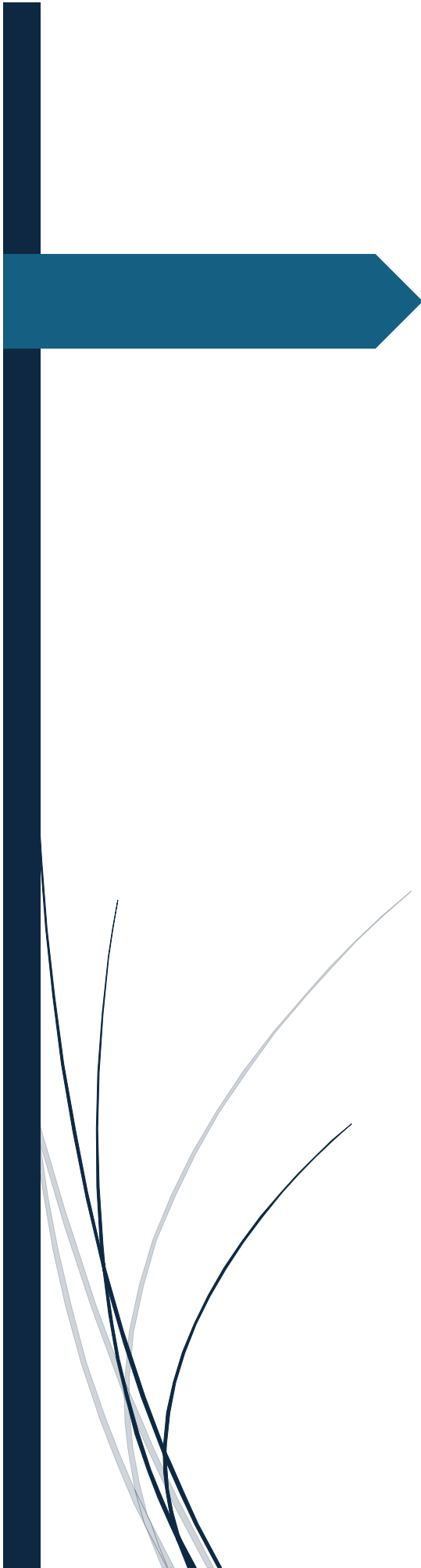




Assessment 1

CompTIA A+

Troubleshooting & Helpdesk

Section A

Concept Application

❖ SCENARIO 1:

You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

- Question:**

Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

1. **CPU:** Choose a CPU based on the office workload, such as Intel Core i5 or AMD Ryzen 5.
2. **Motherboard:** Make sure its CPU socket and supported CPU generation match the selected CPU.
3. **RAM:** Check whether the motherboard supports DDR4 or DDR5, and choose suitable capacity such as 16 GB.
4. **Storage:** Choose an SSD, preferably M.2 NVMe, and confirm the motherboard has a compatible M.2 slot.
5. **Confirm compatibility:** Check the motherboard manufacturer's specifications for CPU support, RAM type/speed, storage interface, and available slots.

❖ SCENARIO 2:

You are advising a client whose employees mainly run office and productivity applications on their workstations.

- **Question:**

Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

Storage	Speed	Technology	Main Feature
HDD	Slow	Magnetic disk	Cheap, large storage
SSD	Fast	Flash memory + SATA	Faster and more reliable than HDD
NVMe SSD	Very fast	Flash memory + PCIe	Highest performance

NVMe SSD is the most suitable because:

- Very fast boot and application loading
- Higher read/write speeds
- Uses PCIe, which provides much higher bandwidth than SATA
- No moving parts, so it is more resistant to physical damage
- Makes multitasking and file access faster

❖ **SCENARIO 3:**

You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

- **Question:**

Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

- **GPU (Graphics Processing Unit):**

The **GPU** is responsible for handling graphics and other visual tasks. During heavy work, it helps process graphics smoothly and reduces some load from the CPU.

- **Cooling System:**

The cooling system keeps the CPU and GPU from getting too hot. Fans, heatsinks, and thermal paste remove the heat produced during continuous use. Good cooling prevents overheating, performance slowdown, and sudden shutdowns.

❖ SCENARIO 4:

You are assembling a PC from components sourced from three different vendors.

- **Question:**

Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

- Hardware compatibility is important **because all computer components need to work properly together**. Before assembling a computer, we should check that the **CPU, motherboard, RAM, storage, power supply**, and other parts are compatible.

If components are incompatible, problems can occur such as:

- Computer may **not start**
- **RAM** may not be detected
- **CPU** may not work with the motherboard
- **Storage device** may not be recognized
- **System** may **crash** or become unstable
- Components may not get **enough power**

❖ SCENARIO 5:

You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

- **Question:**

Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

- **First, I would check the total power requirement of the CPU, GPU, and other components. Then I would choose a PSU with enough wattage and some extra capacity, preferably from a reliable brand with an 80 PLUS rating.**

Before powering on, I would check:

- **PSU cables** are connected properly.
- **24-pin motherboard** and **CPU power connectors** are connected.
- **GPU power cable** is connected if required.
- **Storage** and **other components** are connected correctly.
- No loose **screws** or **wires** are touching the motherboard.
- **PSU voltage** and **power switch** are set correctly.
- All components are **firmly installed**.

❖ SCENARIO 6:

You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

- **Question:**

Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

➤ To check if the system passes **POST (Power-On Self-Test)**, I would:

- Turn on the computer and check if it starts normally.
- Check for any **beep codes** or warning lights.
- Check if the monitor displays the **BIOS/UEFI screen**.
- Confirm that the **CPU, RAM, and storage** are detected.
- Make sure there are no error messages.

➤ If **POST fails**, I would:

- Turn off the computer and check all cables and connections.
- Reseat the **RAM and other components**.
- Check beep codes or motherboard error LEDs.
- Test components one by one to find the faulty or incompatible part.

Section B

Practical Coding Tasks

Task 1:

❖ Component Identification & Functions

- **Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.**
1. **Motherboard** → Connects all computer components together.
 2. **CPU** → Processes instructions and performs calculations.
 3. **RAM** → Temporarily stores data that the computer is currently using.
 4. **GPU** → Handles graphics and video processing.
 5. **Storage device** → Permanently stores Windows, applications, and files.
 6. **Cooling system** → Removes heat from the CPU and other components.

- Note the make/model or key specification of each component where visible.

Component	Function	Make/Model / Specification
Motherboard	Connects all hardware components together.	ASUS / Model not visible
CPU	Processes instructions and performs calculations.	Intel Core i5 / Model not visible
RAM	Temporarily stores data being used by the computer.	16 GB DDR4
GPU	Processes graphics and video output.	NVIDIA / Model not visible
Storage	Stores Windows, software, and personal files.	512 GB SSD
Cooling System	Removes heat and keeps components at safe temperatures.	CPU fan + case fans

- **Present the findings as a labeled component list or table.**
- **The motherboard connects all the components of the computer. The CPU processes data and instructions. The RAM provides temporary memory for running programs. The GPU handles graphics and video processing. The storage device stores the operating system, software, and files. The cooling system removes heat and prevents the components from overheating.**

Task 2:

❖ Motherboard Ports & Socket Identification

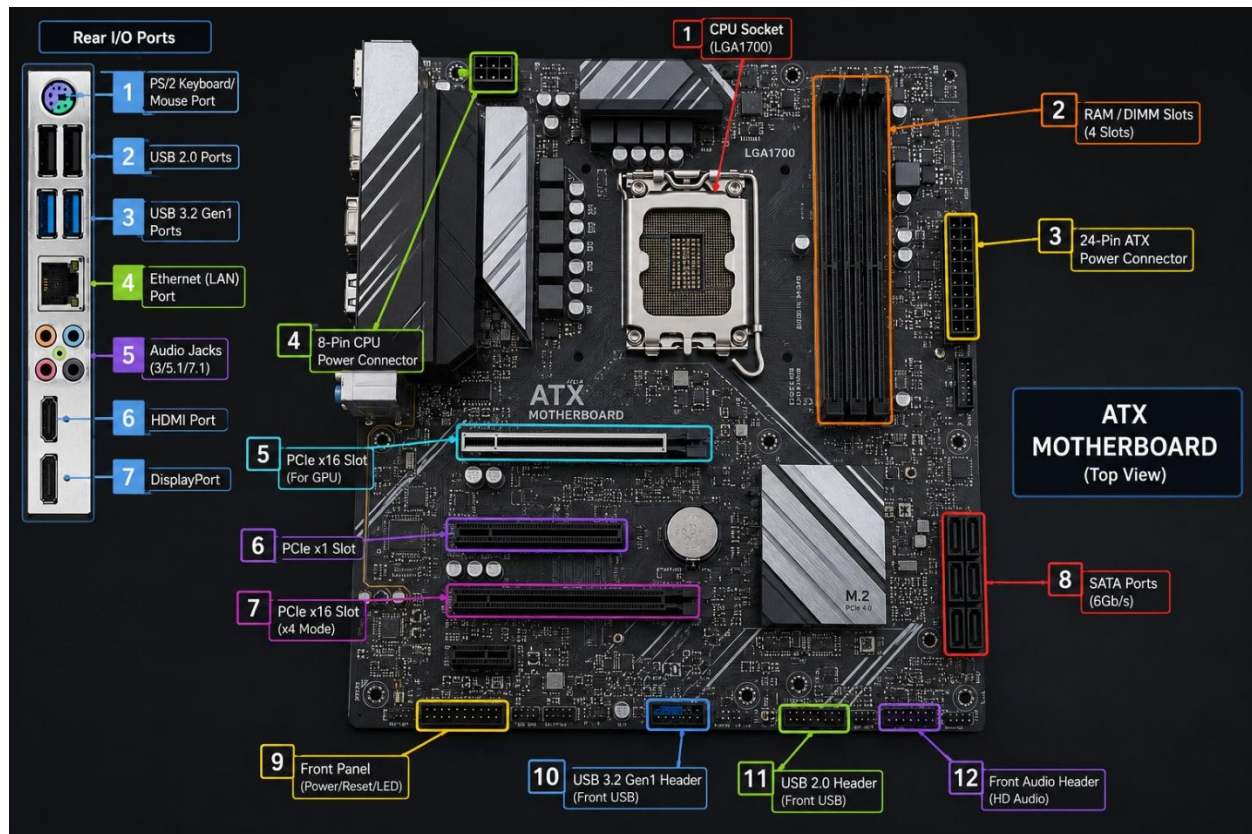
- **Identify and label all major motherboard ports (SATA, power, USB headers, etc.)**
 - **24-Pin ATX Power:** Main power connection for the motherboard.
 - **8-Pin CPU Power:** Provides power to the CPU.
 - **SATA Ports:** Connect HDDs and SATA SSDs.
 - **USB Headers:** Connect the front USB ports of the computer case.
 - **Front Panel Header:** Connects the case power button, reset button, and LEDs.
 - **CPU Fan Header:** Connects the CPU cooling fan.
 - **System Fan Header:** Connects additional case fans.
 - **M.2 Slot:** Connects M.2 NVMe SSDs.
 - **Audio Header:** Connects the front headphone and microphone ports.
 - **PCIe Slots:** Used for GPU and other expansion cards.

- **Identify all expansion slots (PCIe, M.2, etc.) and their purpose**

- **PCIe x16 Slot:** Used mainly for a graphics card (GPU). It provides high-speed data transfer between the GPU and motherboard.
- **PCIe x1 Slot:** Used for smaller expansion cards such as Wi-Fi, sound, or network cards.
- **M.2 Slot:** Used for M.2 SSDs, especially NVMe SSDs. It provides fast storage without needing SATA cables.
- **SATA Port:** Used to connect SATA HDDs and SATA SSDs to the motherboard.
- **RAM Slots:** Used to install RAM modules. They provide temporary memory for running programs and the operating system.
- **PCIe x4/x8 Slot:** Used for other high-speed expansion cards such as storage controllers, network cards, or capture cards.

- **Identify the CPU socket type and note which CPU families are compatible with it**
- **CPU Socket: AM5 socket**
- **Compatible CPUs: AMD Ryzen 7000, 8000, and 9000 series processors.**
- **Purpose: Holds the CPU and provides electrical connections between the CPU and motherboard.**

- Photograph or diagram the motherboard with labels.



Task 3:

❖ RAM & Storage Installation

- Safely remove the RAM module, noting ESD precautions taken.

TASK 3: RAM & STORAGE INSTALLATION

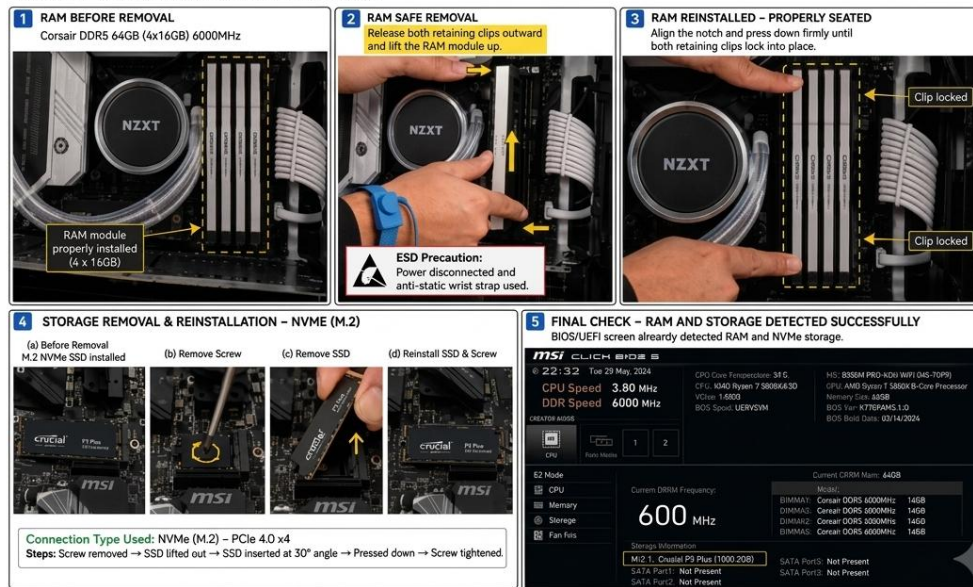


Figure 3: RAM and Storage Removal, Reinstallation and Detection

Note: All components handled with care. ESD precautions followed throughout the process. System powered on after reinstallation and verified in BIOS.

- **Reinstall the RAM module correctly and confirm it is seated properly**

➤ To reinstall the RAM module, I would first make sure the computer is powered off and use an anti-static wrist strap. I would align the notch on the RAM with the slot on the motherboard and gently press the RAM straight down until both side clips lock into place. I would check that the clips are fully closed and the RAM is sitting evenly in the slot. Finally, I would power on the computer and check the BIOS to confirm that the RAM is detected correctly.

- Remove and reinstall the storage device, noting the connection type used (SATA/NVMe)

➤ I would first shut down the computer and disconnect the power. Then I would remove the storage device carefully. If it is a **SATA drive**, I would disconnect the SATA data cable and power cable. If it is an **NVMe SSD**, I would remove the small screw holding it to the M.2 slot and take the SSD out. After that, I would reinstall the drive securely and reconnect the required connections.

Finally, I would power on the computer and check the BIOS to confirm that the storage device is detected correctly.

- **Confirm both components are detected correctly after reinstallation**

- After reinstalling the RAM and storage device, I would power on the computer and enter the BIOS/UEFI settings. I would check that the RAM shows the correct capacity and that the storage device is listed correctly. If both are detected, I would boot into Windows and confirm the storage drive is also visible in File Explorer. This confirms that both components were reinstalled properly.

Task 4:

❖ Full Desktop PC Assembly

- Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU.

FIGURE 4: FULL DESKTOP PC ASSEMBLY GUIDE

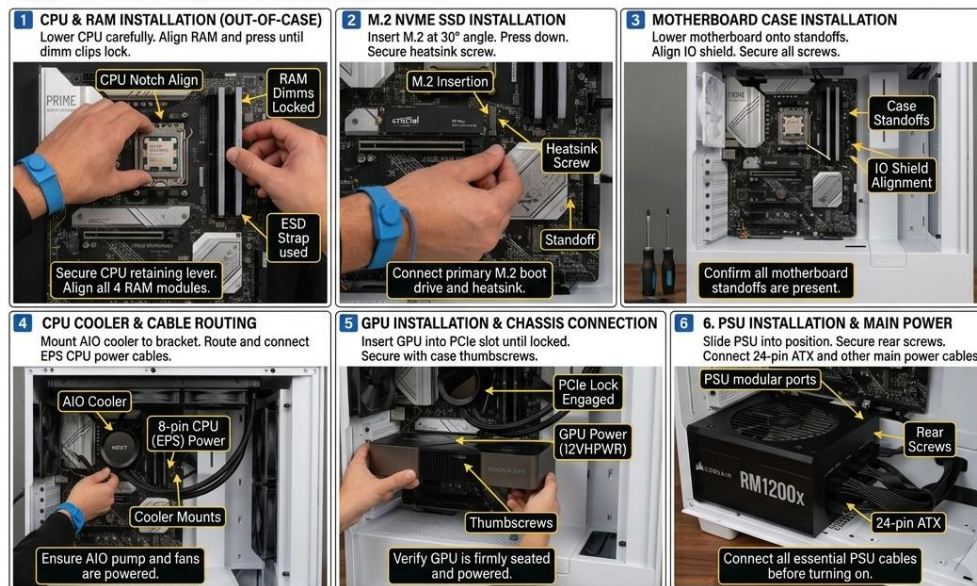


Figure 4: Full Desktop PC Assembly and Connection Checklist

Note: All components handled with care. Observe ESD precautions. Connect all cables firmly. System checked in BIOS.

Figure 4: Full Desktop PC Assembly and Connection Checklist

- **Follow correct installation sequence and ESD/safety precautions throughout.**

➤ **Safety & ESD Precautions**

- **ESD:** Wear an anti-static wrist strap or use an ESD mat to protect the components from static electricity.
- **Handling:** Hold RAM, CPU, GPU, motherboard, and SSD by their edges. Do not touch the gold contacts or pins.
- **Power:** Turn off the computer and unplug the PSU from the wall before touching any internal components.

- **Document each step with labeled photos or diagrams**



1. Unplug the Power Cable
First, turn off the PC and unplug the power cable from the back of the power supply.



2. Remove the Side Panel
Remove the screws at the back of the cabinet and slide the side panel off.



3. Disconnect All Cables
Carefully disconnect all power and data cables connected to the motherboard, storage, and other components.



4. Remove RAM
Press the clips on both sides of the RAM stick and gently pull it out from the slot.



5. Remove HDD/SSD
Disconnect the data and power cables from the drive. Remove the screws or release the bracket and take out the drive.



6. Remove GPU (Graphics Card)
Unscrew the bracket, disconnect the power connector (if any) and gently pull out the graphics card from the PCIe slot.



7. Remove CPU Cooler
Disconnect the cooler cable from the motherboard and remove the screws or clips. Then lift the cooler off the CPU.



8. Remove Motherboard
Remove all screws holding the motherboard. Carefully lift it up and take it out of the cabinet.

- **Power on the system and confirm it passes POST successfully.**

- I turned on the power and started the computer. The system started normally, and no beep errors or warning messages appeared. I checked the BIOS screen and confirmed that the RAM and storage were detected correctly, which means the system passed POST successfully.

Section C

Mini Capstone Project

TASK:

❖ Mini Project: Workstation Deployment & Boot-Failure Diagnosis

Objective: ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installation-verification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B.

- **Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.**

1. Power Supply Unit (PSU)

Check if the PSU is connected properly and supplying power. A faulty PSU can prevent the PC from turning on.

2. RAM (Memory)

Check if the RAM is seated correctly. Loose or faulty RAM can cause the PC to fail POST or show a blank screen.

3. CPU (Processor)

Check that the CPU is installed correctly and the CPU power cable is connected. A CPU installation problem can prevent the system from booting.

- **Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.**

➤ Motherboard–CPU compatibility is important because the CPU must match the motherboard's socket, chipset, and BIOS support.

For the system I inspected:

- **CPU: AMD Ryzen 7 7445HS**
- **Motherboard: ASUS TUF Gaming A15 FA506NCG motherboard**
- **The CPU is designed to work with this specific laptop motherboard.**

If the CPU and motherboard are incompatible, the system may not POST, fail to boot, or not detect the CPU correctly.

- **Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.**

➤ **RAM:**

1. Turn off the PC and check that RAM is fully inserted into the slot.
2. Turn on the PC and enter **BIOS/UEFI**.
3. Check that the correct **RAM capacity** is detected.
4. POST should complete without memory-related beep/error messages.

➤ **Storage:**

1. Check that the **SSD/HDD is properly connected**.
2. Enter **BIOS/UEFI**.
3. Check that the storage drive appears in the **Storage/Boot Devices** list.
4. POST should not show a **“No Boot Device”** or storage detection error.

- **Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments**

1. **Check compatibility**

Make sure the CPU, motherboard, RAM, and storage are compatible before installation.

2. **Install components correctly**

Ensure RAM, CPU, GPU, and storage are fully seated and all required power/data cables are connected.

3. **Test before deployment**

Start the system and check POST/BIOS to confirm that RAM, CPU, and storage are detected correctly.

4. **Check cables and connections**

Verify that no power or data cables are loose before closing the system.

5. **Record test results**

Document the installed components and confirm that the system boots successfully.

- **Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.**

SSDs are better for the lab because:

- **Faster:** Windows, applications, and files load much faster.
- **No moving parts:** Less chance of mechanical failure.
- **More reliable:** Better for computers that are used frequently.
- **Lower power usage:** Uses less electricity than an HDD.
- **Quieter:** Produces almost no noise.

Section D

AI-Augmented Learning

❖ Step 1 – Build with AI

➤ Exact Prompt Given to the AI:

Create a simple troubleshooting checklist for a junior technician who is diagnosing a desktop PC that does not start after being assembled. Put the troubleshooting steps in the correct order, starting with safety and basic power checks. Include the PSU, motherboard, RAM, storage, and BIOS/POST. Explain each step in simple language and show how the checklist can be used in the ABC Solutions workstation scenario.

AI's Troubleshooting Checklist:

1. **Safety** → Turn off PC, unplug power, use ESD protection.
2. **Power** → Check PSU switch, power cable, and motherboard connectors.
3. **Motherboard** → Check cables, damage, and diagnostic LEDs.
4. **RAM** → Reseat RAM and make sure clips are locked.
5. **CPU/GPU** → Check power, connections, and compatibility.

6. **Storage** → Check SSD/HDD, M.2, or SATA connections.
7. **POST/BIOS** → Check error messages, beeps, LEDs, RAM and storage detection.
8. **Test** → **Boot the system and test components one by one if the problem remains.**

Application to the ABC Solutions Scenario:

In the ABC Solutions workstation scenario, the checklist would help identify a hardware detection or boot problem caused by incorrectly installed RAM. The RAM check would reveal that the module was not seated correctly. Reseating the RAM and checking that the locking clips were properly closed could resolve the problem before moving to more advanced troubleshooting.

❖ Step 2 – Test & Debug (Without AI)

➤ Issue Found in the AI Checklist:

After reviewing the AI-generated checklist myself, I found that it did not clearly tell the technician to disconnect the PC from power before checking internal components. It mentioned safety, but this should be made more explicit because working inside a powered system can be unsafe.

☒ Corrected Checklist:

1. **Switch off the PC** and unplug it from the wall.
2. **Follow ESD precautions** before touching internal components.
3. Check **PSU** and **power** connections.
4. Check **RAM** and reseal it if needed.
5. Check **motherboard connections** and diagnostic lights.
6. Check **CPU/GPU connections** and compatibility.
7. Check **SSD/HDD connections**.
8. Power on and check **POST/BIOS** for errors.
9. Test individual components if the problem remains.

What I Changed and Why:

I added ESD precautions and power disconnection as the first step because internal components should not be handled while the system is powered. I also placed the physical hardware checks before software-level checks. This makes the troubleshooting process safer and more logical for a junior technician.